

[24/09/2026 11:04 AM] Imaanaa Harmee:
Introduction to Physiology Practice Qn

Dr. Doc

[24/09/2026 11:04 AM] Imaanaa Harmee: **1. Who is credited with correctly describing the direction of blood circulation and is often called the "father" of physiology?**

- A. Claude Bernard**
- B. Walter Cannon**
- C. William Harvey**
- D. Robert Hooke**
- E. Louis Pasteur**

Answer: C

Explanation: In 1628, **William Harvey** correctly described the direction of blood circulation and is sometimes referred to as the "father" of physiology.

[24/09/2026 11:04 AM] Imaanaa Harmee: **2. Which of the following statements regarding the historical background of physiology is False?**

- A. Claude Bernard described extracellular fluid (ECF) as the "internal environment."**
- B. Walter Cannon termed the maintenance of constant conditions in ECF as homeostasis.**
- C. Physiology is considered an experimental science.**
- D. Claude Bernard coined the term "homeostasis" in the 19th century.**
- E. ECF is also known as milieu interior in French.**

Answer: D

Explanation: While Claude Bernard described the internal environment, it was **Walter Cannon** who actually coined the term "homeostasis".

[24/09/2026 11:04 AM] Imaanaa Harmee: **3. All the following are components of a reflex arc Except:**

- A. Receptors**
- B. Sensory system**
- C. Integrating center**
- D. Endocrine glands**
- E. Effectors**

Answer: D

Explanation: A **reflex arc** is composed specifically of five structures: receptors, sensory system, integrating center, motor system, and effectors. Endocrine glands are part of the hormonal regulatory mechanism.

[24/09/2026 11:04 AM] Imaanaa Harmee: **4. Which of the following is Correct regarding the comparison between nervous and hormonal regulation?**

- A. Hormonal regulation is faster than nervous regulation.**
- B. Nervous effects are mostly localized.**
- C. Both involve chemicals released in large amounts.**
- D. Nervous regulation is faster, while hormonal regulation is slower.**
- E. Neither system works toward homeostasis.**

Answer: D

Explanation: Nervous regulation is faster than hormonal regulation, although both systems work toward the common goal of homeostasis.

[24/09/2026 11:04 AM] Imaanaa Harmee: **5. Which of these physiologic parameters is True for a normal adult?**

- A. Normal pH range: 7.00 – 7.10**
- B. Normal Osmolality: 285 – 300 mosm/L**
- C. Normal body temperature: 38.5 – 39.1°C**
- D. Normal ECF Sodium (Na⁺): 103 meq/L**
- E. Normal ECF Potassium (K⁺): 142 meq/L**

Answer: B

Explanation: The normal osmolality of body fluids is 300 mosm/L, with a range of 285 – 300 mosm/L.

[24/09/2026 11:04 AM] Imaanaa Harmee: **6. All the following are examples of Negative Feedback Mechanisms Except:**

- A. Control of Arterial Blood Pressure (ABP)**
- B. Control of Blood Glucose Level (BGL)**
- C. Control of Body Temperature (BT)**
- D. Blood clotting process**
- E. Regulation of PCO₂ in the blood**

Answer: D

Explanation: Blood clotting is a valuable example of a Positive Feedback Mechanism, whereas most homeostatic parameters like blood pressure and glucose are controlled by Negative Feedback.

[24/09/2026 11:04 AM] Imaanaa Harmee: **7. Which statement about the Positive Feedback Mechanism (PFM) is False?**

- A. It is also known as a "vicious circle."**
- B. Most PFM actions can disturb the internal environment and cause death.**
- C. The LH-surge before ovulation is a harmful example of PFM.**
- D. PFM enhances or repeats the action of the starting stimulus.**
- E. Labor during childbirth is an example of a valuable use of PFM.**

Answer: C

Explanation: While many positive feedback cycles are harmful, the LH-surge, blood clotting, and labor are examples of valuable uses of PFM.

[24/09/2026 11:04 AM] Imaanaa Harmee: **8. Which of the following represents the Correct structural level of organization from simplest to most complex?**

- A. Cell → Tissue → Chemical → Organ → Organism
- B. Chemical → Cell → Tissue → Organ → Organ system → Organism
- C. Organism → Organ system → Organ → Tissue → Cell
- D. Chemical → Tissue → Cell → Organ → Organism
- E. Cell → Organ → Tissue → Chemical → Organism

Answer: B

Explanation: The human body is organized from the **chemical level** to cells, then tissues, organs, organ systems, and finally the complete organism.

[24/09/2026 11:04 AM] Imaanaa Harmee: **9. Which of the following is True regarding Prokaryotes?**

- A. They contain a typical nucleus.
- B. They are generally larger than Eukaryotes.
- C. They possess membrane-bound organelles.
- D. They have a small circular chromosome and no cytoskeleton.
- E. RNA synthesis occurs in the nucleus and protein synthesis in the cytoplasm.

Answer: D

Explanation: Prokaryotes (like bacteria) are characterized by having no nucleus, no cytoskeleton, and a small circular chromosome.

[24/09/2026 11:04 AM] Imaanaa Harmee: **10. The cell membrane is entirely composed of the following Except:**

- A. Proteins (55%)
- B. Phospholipids (25%)
- C. Cholesterol (13%)
- D. Water (10%)
- E. Carbohydrates (3%)

Answer: D

Explanation: The plasma membrane is composed of **proteins (55%)**, lipids (42% total, including phospholipids and cholesterol), and carbohydrates (3%).

[24/09/2026 11:04 AM] Imaanaa Harmee: **11. Which statement regarding the lipid bilayer of the cell membrane is Correct?**

- A. Hydrophilic tails face each other in the interior.
- B. The lipid bilayer is impermeable to water-soluble substances like ions and glucose.
- C. Fat-soluble substances like O₂ and CO₂ cannot diffuse through it.
- D. Hydrophobic ends face the ECF and ICF.
- E. It is a rigid, solid structure.

Answer: B

Explanation: The **lipid bilayer** is impermeable to water and water-soluble substances (ions, glucose), but allows fat-soluble substances (O₂, CO₂, alcohol) to diffuse through.

[24/09/2026 11:04 AM] Imaanaa Harmee: **12. Which organelle is correctly matched with its function?**

- A. Ribosomes: Lipid biosynthesis**
- B. Lysosomes: Power plant of the cell**
- C. Golgi Complex: Modifying and packaging secretory granules**
- D. Mitochondria: Intracellular digestive system**
- E. Smooth ER: Site of protein synthesis**

Answer: C

Explanation: The **Golgi complex** modifies and packages substances into secretory granules. Ribosomes/Rough ER are for protein synthesis, Mitochondria are power plants, and Lysosomes are for digestion.

[24/09/2026 11:04 AM] Imaanaa Harmee: **13. All the following are functions of the Smooth Endoplasmic Reticulum (sER) Except:**

- A. Glycogen storage in the liver**
- B. Calcium storage in muscle cells**
- C. Lipid biosynthesis**
- D. Drug metabolism (detoxifying)**
- E. Site of rRNA synthesis**

Answer: E

Explanation: The **sER** is involved in glycogen storage, calcium storage, lipid biosynthesis, and drug metabolism. The nucleolus is the site for rRNA synthesis.

[24/09/2026 11:04 AM] Imaanaa Harmee: **14. Which statement about nucleotides is False?**

- A. They are the building units of DNA and RNA.**
- B. They can act as high-energy molecules like ATP.**
- C. In RNA, the base Thymine (T) is used instead of Uracil (U).**
- D. They act as coenzymes (NAD, FAD).**
- E. They can regulate chemical reactions (e.g., cAMP).**

Answer: C

Explanation: In **RNA**, the pyrimidine base Thymine (T) is replaced by Uracil (U).

[24/09/2026 11:04 AM] Imaanaa Harmee: **15. Which transport mechanism moves substances "uphill" against an electrochemical gradient and consumes energy?**

- A. Simple diffusion**
- B. Facilitated diffusion**
- C. Osmosis**

- D. Active transport**
- E. Passive transport**

Answer: D

Explanation: **Active transport** involves moving substances against a concentration or electrochemical gradient, which consumes energy.

[24/09/2026 11:04 AM] Imaanaa Harmee: **16. Which of the following is Correct regarding the Na⁺-K⁺ ATPase pump?**

- A. It pumps 2 Na⁺ outward and 3 K⁺ inward.**
- B. It consumes diffusion energy rather than ATP.**
- C. It pumps 3 Na⁺ outward and 2 K⁺ inward.**
- D. It maintains a high concentration of Sodium inside the cell.**
- E. It is a form of secondary active transport.**

Answer: C

Explanation: The **Na-K pump** is a primary active transport mechanism that pumps 3 Na⁺ outward and 2 K⁺ inward using energy from ATP.

[24/09/2026 11:04 AM] Imaanaa Harmee: **17. Regarding cell junctions, which statement is True?**

- A. Tight junctions are used for cell-cell communication.**
- B. Gap junctions maintain tissue integrity against mechanical stress.**
- C. Desmosomes are abundant in tissues needing to withstand mechanical stress, like the skin.**
- D. Adherens junctions regulate paracellular permeability.**
- E. Connexins are the primary proteins in tight junctions.**

Answer: C

Explanation: **Desmosomes** provide strong intercellular adhesion to withstand mechanical stress and are abundant in the skin and heart .

[24/09/2026 11:04 AM] Imaanaa Harmee: **18. If a cell is placed in a solution and it swells, the solution is said to be:**

- A. Isotonic**
- B. Hypotonic**
- C. Hypertonic**
- D. Iso-osmotic**
- E. Concentrated**

Answer: B

Explanation: A **hypotonic solution** has a lower concentration of solutes than the cell, causing water to diffuse into the cell, which leads to swelling.

[24/09/2026 11:04 AM] Imaanaa Harmee: **19. Which of the following constitutes the largest percentage of an average adult human's body weight?**

- A. Proteins**
- B. Fats**
- C. Water**
- D. Minerals**
- E. Carbohydrates**

Answer: C

Explanation: **Water** constitutes approximately 60% of the total body weight in an average adult human .

[24/09/2026 11:04 AM] Imaanaa Harmee: **20. Which of the following statements about body fluid compartments is False?**

- A. ECF is subdivided into interstitial fluid and blood plasma.**
- B. ICF makes up one-third of the total body water.**
- C. Plasma contains a higher protein concentration than interstitial fluid.**
- D. Transcellular fluids include CSF and synovial fluid.**
- E. A lean person has a higher percentage of body water than an obese person.**

Answer: B

Explanation: **Intracellular fluid (ICF)** actually makes up two-thirds of the total body water, while ECF makes up one-third.

[24/09/2026 11:04 AM] Imaanaa Harmee:
Respiratory system Physiology Practice Qn

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[24/09/2026 11:04 AM] Imaanaa Harmee: **1. In accordance with Boyle's Law and the mechanics of breathing, which of the following sequences Correctly describes the events leading to inspiration?**

- A. Diaphragm relaxes → Thoracic volume increases → Alveolar pressure rises → Air flows out.**
- B. Diaphragm contracts → Thoracic volume decreases → Alveolar pressure drops → Air flows in.**
- C. Diaphragm contracts → Thoracic volume increases → Alveolar pressure becomes sub-atmospheric → Air flows in.**
- D. External intercostals relax → Thoracic volume increases → Intrapleural pressure becomes less negative → Air flows in.**
- E. Abdominal muscles contract → Thoracic volume increases → Alveolar pressure drops → Air flows in.**

Answer: C

Explanation: Inspiration is an active process initiated by the contraction of the diaphragm and external intercostal muscles. According to Boyle's Law, as volume increases, pressure must decrease. The contraction of these muscles increases thoracic cavity volume, which

causes alveolar (intrapulmonary) pressure to fall below atmospheric pressure (sub-atmospheric), creating the pressure gradient necessary for air to flow into the lungs.

[24/09/2026 11:04 AM] Imaanaa Harmee: **2. Applying the Law of Laplace ($P=2T/r$) to the alveoli, which of the following is the Best answer regarding the role of surfactant?**

- A. It increases surface tension in large alveoli to ensure they do not over-expand.**
- B. It ensures that the pressure in smaller alveoli is higher than in larger ones.**
- C. It reduces surface tension more effectively in smaller alveoli, preventing them from collapsing into larger ones.**
- D. It increases the radius of the alveolus to decrease the pressure required for inflation.**
- E. It is composed primarily of carbohydrates that increase lung recoil forces.**

Answer: C

Explanation: Without surfactant, the Law of Laplace dictates that smaller alveoli (smaller radius r) would have much higher pressure (P) and would empty their air into larger alveoli, leading to collapse (atelectasis). Surfactant, primarily composed of the phospholipid **DPPE**, reduces surface tension dramatically, which stabilizes the alveoli, increases distensibility, and reduces the work of breathing.

[24/09/2026 11:04 AM] Imaanaa Harmee: **3. All of the following factors contribute to Reduced Lung Compliance EXCEPT:**

- A. Pulmonary fibrosis (increase of fibrous tissue).**
- B. Alveolar edema, which prevents the inflation of some alveoli.**
- C. Increased pulmonary venous pressure leading to lung engorgement with blood.**
- D. Pulmonary emphysema, which results in the destruction of alveolar walls.**
- E. A developmental insufficiency of surfactant production in premature infants.**

Answer: D

Explanation: Compliance is the extent to which lungs expand per unit increase in pressure. Fibrosis, edema, and venous congestion all make the lungs "stiffer" and harder to expand, thereby **reducing** compliance. Conversely, emphysema and the normal aging lung actually result in increased compliance because the destruction of alveolar walls reduces the elastic recoil of the lung.

[24/09/2026 11:04 AM] Imaanaa Harmee: **4. Which of the following lung volumes or capacities CANNOT be measured directly by simple Spirometry?**

- A. Tidal Volume (V_T) and Inspiratory Capacity (IC).**
- B. Vital Capacity (VC) and Expiratory Reserve Volume (ERV).**
- C. Residual Volume (RV) and Functional Residual Capacity (FRC).**
- D. Inspiratory Reserve Volume (IRV) and Tidal Volume (V_T).**
- E. Minute Ventilation and Vital Capacity (VC).**

Answer: C

Explanation: Spirometry can only measure air that can be moved into or out of the lungs. Because the lungs cannot be completely emptied even after a forced expiration, the **Residual Volume (RV)** cannot be measured directly. Consequently, any capacity that includes RV—

such as Functional Residual Capacity (FRC) and Total Lung Capacity (TLC)—must be measured indirectly using techniques like helium dilution.

[24/09/2026 11:04 AM] Imaanaa Harmee: **5. In a healthy adult male with a Tidal Volume of 500 ml, a Respiratory Rate of 12 breaths/min, and an Anatomical Dead Space of 150 ml, which statement is Correct?**

- A. The Minute Ventilation is 4200 ml/min.**
- B. The Alveolar Ventilation per minute is 4200 ml/min.**
- C. Increasing the respiratory rate is always more effective for gas exchange than increasing tidal volume.**
- D. The physiologic dead space is always smaller than the anatomic dead space in diseased lungs.**
- E. All 500 ml of each breath reaches the gas exchange areas (respiratory zone).**

Answer: B

Explanation: Alveolar ventilation (V_A) is calculated as $\text{Freq} \times (V_T - V_D)$. In this case: $12 \times (500 - 150) = 4200$ ml/min. Minute ventilation would be $12 \times 500 = 6000$ ml/min. Increasing tidal volume is generally more effective than increasing rate because it reduces the proportion of the breath wasted in the dead space.

[24/09/2026 11:04 AM] Imaanaa Harmee: **6. Identify the False statement regarding the Muscles of Respiration:**

- A. The diaphragm is the primary muscle of quiet inspiration and is supplied by the phrenic nerves.**
- B. External intercostals pull the ribs upward and forward to increase the anteroposterior diameter.**
- C. Quiet expiration is normally a passive process caused by the elastic recoil of the lungs.**
- D. Forced expiration is achieved by the contraction of the sternocleidomastoid and scalene muscles.**
- E. The rectus abdominis and internal intercostals are major muscles used in forced expiration.**

Answer: D

Explanation: The sternocleidomastoid and scalene muscles are actually **accessory muscles of inspiration** used to further elevate the rib cage. Forced expiration is achieved by the abdominal muscles (which push the diaphragm up) and the internal intercostal muscles (which depress the ribs).

[24/09/2026 11:04 AM] Imaanaa Harmee: **7. The "Work" of breathing is divided into three fractions. Which of the following is the Correct definition for one of these fractions?**

- A. Viscosity work: The energy required to move air through the tracheobronchial tree.**
- B. Compliance (Elastic) work: The work required to expand the lungs against the lung and chest elastic forces.**
- C. Airway resistance work: The work required to overcome the sticky resistance of the**

lung tissues.

D. Surface tension work: The energy used to increase the production of DPPC by Type II cells.

E. Active expiration work: The work done by the diaphragm to push air out during quiet breathing.

Answer: B

Explanation: The three fractions are: (1) **Compliance/Elastic work** (overcoming elastic forces), (2) Viscosity work (overcoming tissue resistance), and (3) Airway resistance work (moving air through passages). During quiet breathing, only 3–5% of body energy is used for this work.

[24/09/2026 11:04 AM] Imaanaa Harmee: **8. In the context of Ventilation Rates, which of the following statements is False?**

A. Minute Ventilation is the product of Tidal Volume and Respiratory Rate.

B. Alveolar Ventilation is the rate at which new air reaches the respiratory zone.

C. Anatomic dead space refers to air filling passages like the nose and trachea where gas exchange does not occur.

D. Increasing respiratory frequency is always more effective for increasing alveolar ventilation than increasing tidal volume.

E. Physiologic dead space includes both anatomic dead space and non-functional alveoli.

Answer: D

Explanation: While increasing either tidal volume (V_T) or frequency can raise alveolar ventilation, **increasing V_T** is often more effective. This is because a larger tidal volume reduces the proportion of each breath that is wasted in the anatomic dead space, ensuring a higher volume of fresh air actually reaches the alveoli.

[24/09/2026 11:09 AM] Imaanaa Harmee: **9. Identify the False statement regarding the Muscles of Respiration:**

A. The diaphragm is the primary muscle of quiet inspiration and is supplied by the phrenic nerves.

B. External intercostals pull the ribs upward and forward to increase the anteroposterior diameter.

C. Quiet expiration is normally a passive process caused by the elastic recoil of the lungs.

D. Forced expiration is achieved by the contraction of the sternocleidomastoid and scalene muscles.

E. The rectus abdominis and internal intercostals are major muscles used in forced expiration.

Answer: D

Explanation: The sternocleidomastoid and scalene muscles are actually **accessory muscles of inspiration** used to further elevate the rib cage. Forced expiration is achieved by the abdominal muscles (which push the diaphragm up) and the internal intercostal muscles (which depress the ribs).

[24/09/2026 11:09 AM] Imaanaa Harmee: **10.Regarding Pulmonary and Bronchial Circulation, all the following are Correct EXCEPT:**

- A. Pulmonary circulation is a low-pressure, high-flow system.**
- B. The bronchial circulation serves the respiratory zone where gas exchange occurs.**
- C. Bronchial circulation constitutes about 1–2% of the total cardiac output.**
- D. Pulmonary arterial branches have larger diameters than their systemic counterparts.**
- E. Half of the bronchial blood flow drains into the pulmonary circulation, adding deoxygenated blood to oxygenated blood.**

Answer: B

Explanation: The bronchial circulation supplies systemic arterial blood to the trachea, bronchial tree, and supporting tissues, but it **does not extend to the respiratory zone**. The respiratory zone is served exclusively by the pulmonary circulation for the purpose of gas exchange.

[24/09/2026 11:09 AM] Imaanaa Harmee: **11.Which of the following is Not correct regarding Henry's Law and gas solubility?**

- A. Dissolved gas molecules move randomly and possess kinetic energy.**
- B. Partial pressure is determined by concentration divided by the solubility coefficient.**
- C. Oxygen is 20 times more soluble in body fluids than carbon dioxide.**
- D. For a given concentration, the partial pressure of CO₂ is less than 1/20th that of O₂ .**
- E. The solubility coefficient is defined at a specified pressure and temperature**

Answer: C

[24/09/2026 11:09 AM] Imaanaa Harmee: **12.Which of the following represents a Physiological Shunt situation regarding the Ventilation-Perfusion Ratio (VA /Q)?**

- A. VA /Q equals infinity.**
- B. Perfusion is zero, but ventilation is adequate.**
- C. VA /Q is below normal, resulting in a fraction of venous blood failing to become oxygenated.**
- D. Alveolar air becomes equal to humidified inspired air.**
- E. This condition is most common at the apex of the lung in an upright person.**

Answer: C

Explanation: A **physiological shunt** occurs whenever VA /Q is below normal, meaning there is inadequate ventilation to provide the oxygen needed to oxygenate the blood flowing through the capillaries. This "shunted blood" passes through the lungs without being oxygenated. Option A and B describe physiological dead space (VA /Q=infinity). Option E is incorrect because shunts are more common at the base of the lung, whereas dead space is more common at the apex.

[24/09/2026 11:09 AM] Imaanaa Harmee: **13.All of the following factors will shift the oxygen-hemoglobin dissociation curve to the right, thereby reducing hemoglobin's**

affinity for oxygen, EXCEPT:

- A. An increase in blood temperature.
- B. An increase in the concentration of 2,3-diphosphoglycerate (DPG).
- C. A decrease in the partial pressure of carbon dioxide (PCO_2).
- D. An increase in hydrogen ion concentration (decrease in pH).
- E. Strenuous physical exercise involving increased muscle metabolism.

Answer: C

Explanation: A decrease in PCO_2 actually shifts the curve to the left, meaning hemoglobin binds oxygen more tightly (increased affinity). Increases in H^+ ions, PCO_2 , temperature, and 2,3-DPG all shift the curve to the right to facilitate the unloading of oxygen to active tissues.

[24/09/2026 11:09 AM] Imaanaa Harmee: **14.Regarding the Bohr Effect, identify the True statement:**

- A. It describes how the pickup of CO_2 at the lungs facilitates the release of oxygen.
- B. It is quantitatively more important for CO_2 transport than the Haldane effect.
- C. It involves a rightward and downward shift of the curve in response to increased blood CO_2 and H^+ ions.
- D. It occurs primarily in the lungs to help hemoglobin bind more oxygen.
- E. It is caused by the competitive binding of Nitrogen at high altitudes.

Answer: C

Explanation: The Bohr Effect describes the shift of the O_2 dissociation curve to the **right** in peripheral tissues where CO_2 and H^+ levels are high. This shift "forces" oxygen away from hemoglobin, increasing delivery to the tissues. The Haldane effect is actually more important for CO_2 transport than the Bohr effect is for O_2 transport.

[24/09/2026 11:09 AM] Imaanaa Harmee: **15.In the context of Acclimatization to high altitudes, which of the following occurs within 2 to 3 days?**

- A. The respiratory center becomes seven times more sensitive to PCO_2 .
- B. Hemoglobin affinity for oxygen increases by 500% to compensate for thin air.
- C. The respiratory center loses about four-fifths of its sensitivity to changes in PCO_2 and hydrogen ions.
- D. Peripheral chemoreceptors stop functioning to prevent over-ventilation.
- E. The oxygen-hemoglobin dissociation curve shifts permanently to the left.

Answer: C

Explanation: Acclimatization allows mountain climbers to withstand low O_2 because the brain's respiratory center becomes less sensitive to PCO_2 and H^+ changes. This prevents the "ventilatory blow-off" of CO_2 from inhibiting the necessary increase in respiration driven by low oxygen.

[24/09/2026 11:09 AM] Imaanaa Harmee: **16.Which of the following is the Correct**

mechanism by which Central Chemoreceptors are stimulated?

A. Hydrogen ions (H^+) in the blood cross the blood-brain barrier to directly stimulate the neurons.

B. Oxygen levels in the cerebrospinal fluid (CSF) trigger the chemosensitive area.

C. Blood CO_2 crosses the blood-brain barrier, reacts with water to form H^+ , which then stimulates the receptors.

D. Carbon monoxide (CO) binds to the receptors to increase respiratory rate during exercise.

E. Bicarbonate ions (HCO_3^-) act as the primary stimulus for the medullary neurons.

Answer: C

Explanation: Although H^+ is the primary stimulus for these neurons, H^+ ions cannot easily cross the blood-brain barrier. However, CO_2 passes through the barrier almost as if it does not exist. Once across, it reacts with water to form new H^+ ions in the CSF and interstitial fluid, which then excite the chemosensitive neurons.